

ITA0609—MACHINE LEARNING

K. SRINIVASA REDDY (192525452)

```
▶ import pandas as pd

data = {
    "Name": ["Arun", "Priya", "Rahul", "Sneha"],
    "Marks": [85, 65, 90, 70]
}

df = pd.DataFrame(data)

print("Original Data:")
print(df)

# Filter marks greater than 75
filtered = df[df["Marks"] > 75]

print("\nFiltered Data:")
print(filtered)
```

```
... Original Data:
      Name  Marks
0   Arun     85
1  Priya     65
2  Rahul     90
3  Sneha     70
```

```
Filtered Data:
      Name  Marks
0   Arun     85
2  Rahul     90
```



```
import pandas as pd
```

```
data = {  
    "Name": ["Arun", "Priya", "Rahul", "Sneha", "Kiran"],  
    "Marks": [85, 65, 90, 75, 88],  
    "Attendance": [90, 80, 95, 70, 85]  
}
```

```
df = pd.DataFrame(data)
```

```
# Multiple conditions
```

```
result = df[(df["Marks"] > 80) & (df["Attendance"] > 85)]
```

```
print(result)
```

```
...      Name  Marks  Attendance  
0    Arun      85           90  
2    Rahul      90           95
```

```
import pandas as pd
```

```
data = {  
    "Name": ["Arun", "Priya", "Rahul", "Sneha"],  
    "Marks": [85, 65, 90, 70]  
}
```

```
df = pd.DataFrame(data)
```

```
# Sort Marks ascending and Name descending
```

```
result = df.sort_values(  
    by=["Marks", "Name"],  
    ascending=[True, False]  
)
```

```
print(result)
```

```
      Name  Marks  
1  Priya      65  
3  Sneha      70  
0  Arun      85  
2  Rahul      90
```

```
import pandas as pd

data = {
    "Name": ["Arun", "Priya", "Arun", "Rahul"],
    "Marks": [85, 90, 85, 75]
}

df = pd.DataFrame(data)

# Remove duplicate rows
result = df.drop_duplicates()

print(result)
```

	Name	Marks
0	Arun	85
1	Priya	90
3	Rahul	75

```
import pandas as pd

data = {
    "Department": ["CSE", "CSE", "ECE", "ECE"],
    "Marks": [80, 90, 70, 85]
}

df = pd.DataFrame(data)

# Aggregate data
result = df.groupby("Department")["Marks"].agg(["sum", "mean", "max"])

print(result)
```

	sum	mean	max
Department			
CSE	170	85.0	90
ECE	155	77.5	85

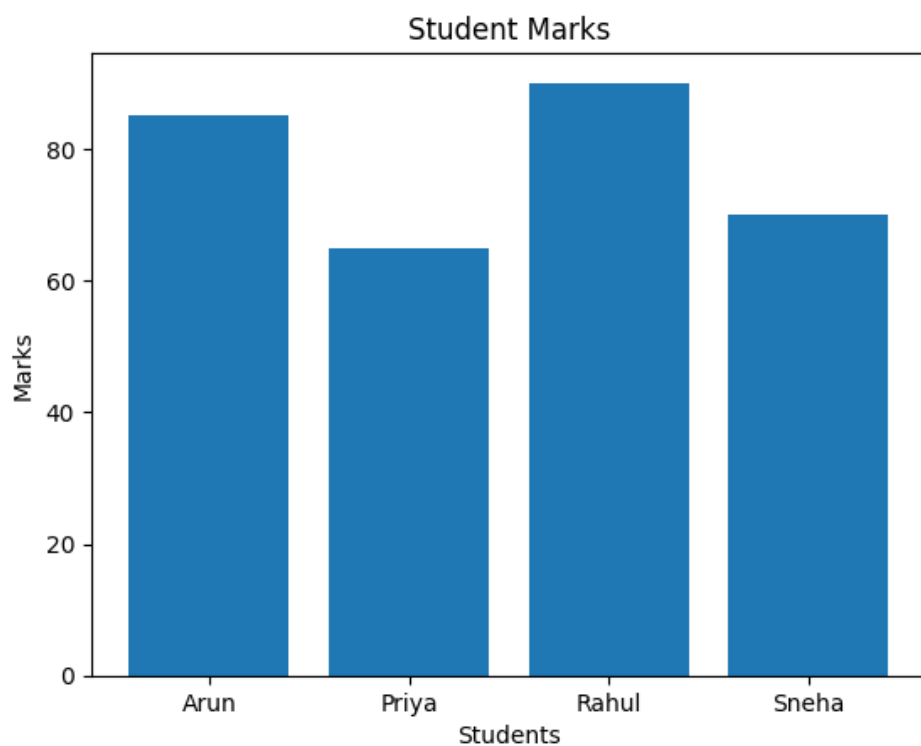
```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["Arun", "Priya", "Rahul", "Sneha"],
    "Marks": [85, 65, 90, 70]
}

df = pd.DataFrame(data)

plt.bar(df["Name"], df["Marks"])
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

...

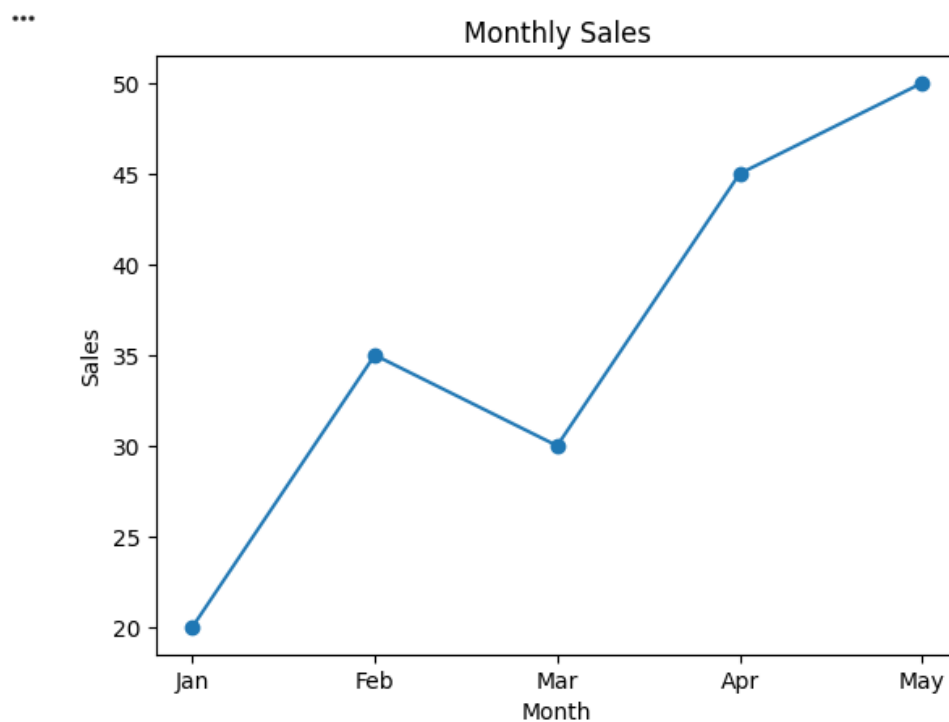


```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Month": ["Jan", "Feb", "Mar", "Apr", "May"],
    "Sales": [20, 35, 30, 45, 50]
}

df = pd.DataFrame(data)

plt.plot(df["Month"], df["Sales"], marker="o")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales")
plt.show()
```

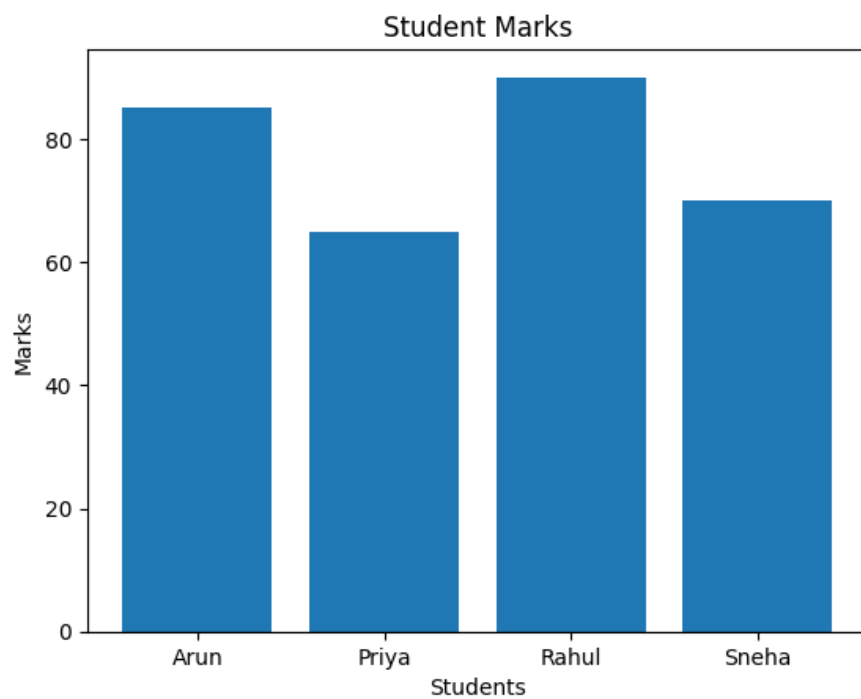


```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Name": ["Arun", "Priya", "Rahul", "Sneha"],
    "Marks": [85, 65, 90, 70]
}

df = pd.DataFrame(data)

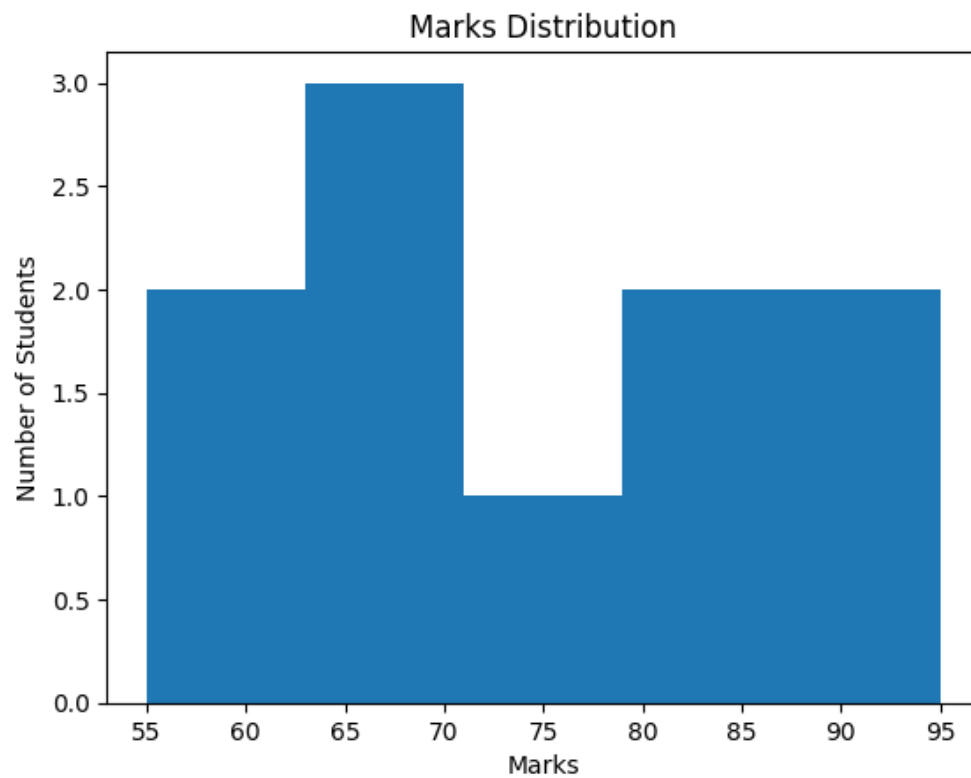
plt.bar(df["Name"], df["Marks"])
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```



```
import matplotlib.pyplot as plt

marks = [55, 60, 65, 70, 70, 75, 80, 85, 90, 95]

plt.hist(marks, bins=5)
plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Marks Distribution")
plt.show()
```

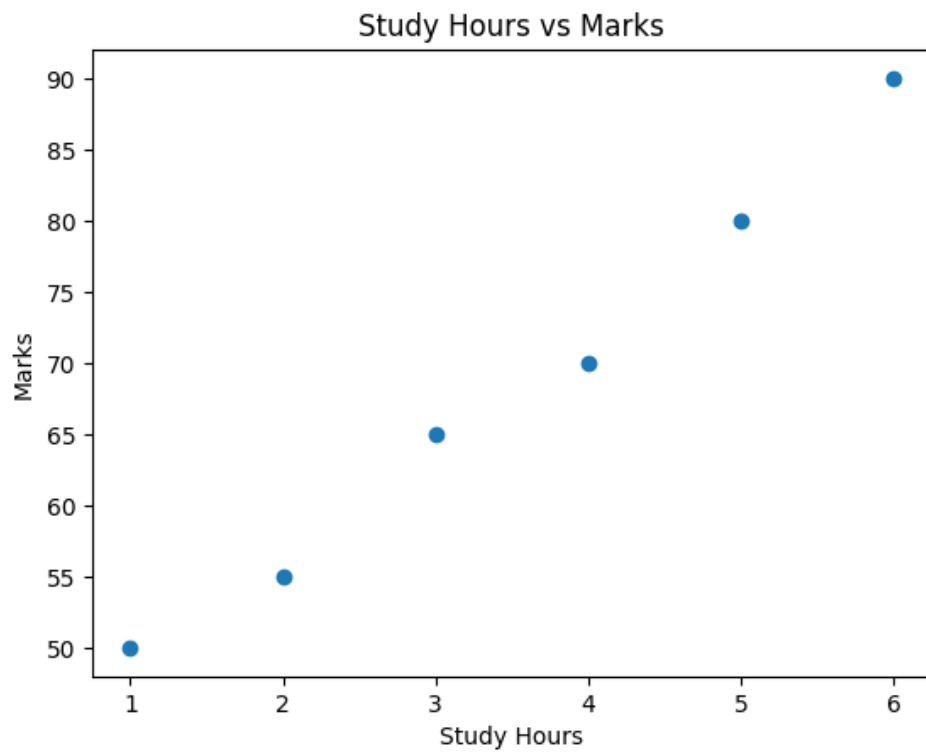


```
▶ import matplotlib.pyplot as plt

hours = [1, 2, 3, 4, 5, 6]
marks = [50, 55, 65, 70, 80, 90]

plt.scatter(hours, marks)
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Study Hours vs Marks")
plt.show()
```

...



```
import matplotlib.pyplot as plt

subjects = ["Maths", "Python", "DBMS", "Networks"]
marks = [30, 25, 20, 25]

plt.pie(marks, labels=subjects, autopct="%1.1f%%")
plt.title("Subject Marks Distribution")
plt.show()
```

